

16. $C_n = C_{n-1} + C_{n-2}$
 Let $F_n = \begin{vmatrix} 1 & -1 & 0 & \dots & 0 \\ 0 & 1 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 1 \end{vmatrix}$

$C_{11} = (-1)^2 (F_{n-1}) = F_{n-1}$

$C_{12} = (-1)^3 \begin{vmatrix} 1 & -1 \\ 0 & F_{n-2} \end{vmatrix}$

$\Rightarrow$ Since $|0 \ F_{n-2}|$ is in

C form $\Rightarrow \det |0 \ F_{n-2}| = (1)(F_{n-2})$

$\Rightarrow C_{12} = -F_{n-2}$

$\Rightarrow F_n = C_{11}C_{11} + C_{12}C_{12}$
 $= F_{n-1} + F_{n-2}$

40. $\begin{vmatrix} 2 & -1 & 0 & 0 & 0 \\ 0 & 2 & -1 & 0 & 0 \\ 0 & 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{vmatrix}$

My guess would be to eliminate all rows/cols found by the 2×2 matrix and calculate the remaining 3×3 . Repeat this for all "detectors" 2×2 matrices

$\begin{vmatrix} 2 & -1 & 0 & 0 & 0 \\ 0 & 2 & -1 & 0 & 0 \\ 0 & 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{vmatrix} \rightarrow \begin{vmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{vmatrix}$
 $= 2^5 = 32$

6.1. $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$ b. $\det(B) = \det(B^T) = 0$

$\Rightarrow \det(B^T B) = \det(B) \det(B^T) = 0$
 $= 0$

a. $\lambda = 0 \Rightarrow B$ is singular c. unknown since B^T and B may have different

$b \times$ has 2 nonzero solutions $\Rightarrow x^2 = 2$ present
 $\therefore \text{Rank}(B) = 2$

d. $B^2 \Rightarrow x^2$, meaning $x^2 = 0, 1, 4$

$(B^2 = 2)$ yields three to have $\Rightarrow$ values: 1, 2, 5

$(B^2 = 5) \Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$

34. $X = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}$
 $\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} X^{-1} = \begin{bmatrix} y_1^T & \dots & y_n^T \end{bmatrix}$

$\Rightarrow X \Lambda X^{-1} = \begin{bmatrix} x_1 \lambda_1 & \dots & x_n \lambda_n \end{bmatrix} X^{-1}$

$\Rightarrow \begin{bmatrix} x_1 \lambda_1 y_1^T & \dots & x_n \lambda_n y_n^T \end{bmatrix}$

32. $S_n = F_{2n+2}$
 $F_k = F_{k-1} + F_{k-2}; k \geq 2n+2$
 $F_{2n+2} = F_{2n+1} + F_{2n}$
 $= (F_{2n} + F_{2n-1}) + F_{2n}$
 $= 2F_{2n} + (F_{2n} - F_{2n-2})$
 $= 3F_{2n} - F_{2n-2}$

33. $A = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \rightarrow A C^T = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = 3I$

$C_{21} = -3, C_{22} = 1, C_{23} = 1$
 $C_{31} = -6, C_{32} = -2, C_{33} = 1$

$\Rightarrow C = \begin{bmatrix} 6 & -3 & 0 \\ -3 & 1 & 1 \\ -6 & -2 & 1 \end{bmatrix}$

Changing the 4 dots but affect the identity matrix as it always remains 0: $C_{13} = 0$

41. $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$

a. First: $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$

for all 2×2 determinants, simply copy the b-c elements

$\Rightarrow \begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 2 + 1 + 1 = 4$

29. a. $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 6 \end{bmatrix}$ b. $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 6 \end{bmatrix}$ c. $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 6 \end{bmatrix}$

$\det(A) = 6$
 $\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 6-\lambda \end{bmatrix} = (1-\lambda)(1-\lambda)(6-\lambda) = 0$

$\Rightarrow \lambda = 1, 1, 6$

62. $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
 $\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 0 \\ 0 & 2-\lambda \end{bmatrix} = (1-\lambda)(2-\lambda) = 0$

$\Rightarrow \lambda = 1, 2$

6.1. $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
 $\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 0 \\ 0 & 2-\lambda \end{bmatrix} = (1-\lambda)(2-\lambda) = 0$

$\Rightarrow \lambda = 1, 2$

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33. If you subtract one from the least entry, elimination produces a row full of zeroes. As such, the matrix is no longer invertible.
 $\therefore \det \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{bmatrix} = 0$

26. $A = \begin{bmatrix} x_p & x_d & x_b \\ y_p & y_d & y_b \\ z_p & z_d & z_b \end{bmatrix}$

$\begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \sin \theta \sin \phi & \sin \theta \cos \phi & \cos \theta \\ \cos \theta & -\sin \theta & 0 \end{bmatrix}$

$\det A = (\sin^2 \theta \cos^2 \phi - \sin^2 \theta \sin^2 \phi) - \cos^2 \theta$

$= \sin^2 \theta (\cos^2 \phi - \sin^2 \phi) - \cos^2 \theta$

$= \sin^2 \theta \cos 2\phi - \cos^2 \theta$

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